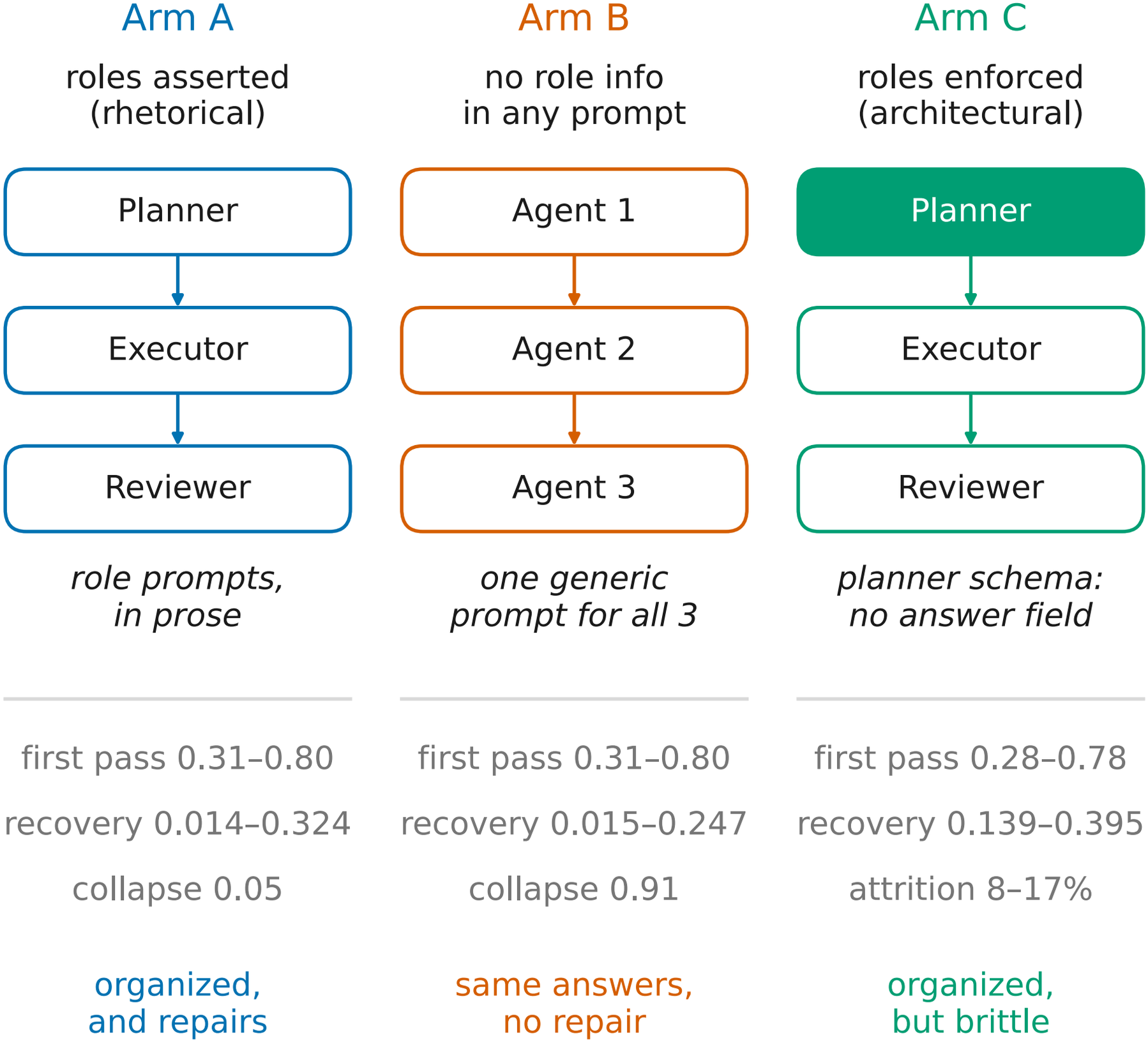
Arm A
roles asserted
(rhetorical)
Planner
Executor
Reviewer
role prompts,
in prose
first pass 0.31–0.80
recovery 0.014–0.324
collapse 0.05
organized,
and repairs
Arm B
no role info
in any prompt
Agent 1
Agent 2
Agent 3
one generic
prompt for all 3
first pass 0.31–0.80
recovery 0.015–0.247
collapse 0.91
same answers,
no repair
Arm C
roles enforced
(architectural)
Planner
Executor
Reviewer
planner schema:
no answer field
first pass 0.28–0.78
recovery 0.139–0.395
attrition 8–17%
organized,
but brittle
Identical in all arms: topology, routing and budget
(3×2=6 cycles, max_evaluator_calls=6, temperature=0).
Only the prompt or schema differs.
Ranges span the three actors; ‘collapse’ is the
mechanical functional-collapse rate (no judge).